

Shubhangi Agarwal

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SUMMARY

I am a researcher with an interest in **Data Mining, Graphs, Artificial Intelligence** and **Databases**. Currently, I am working on 'BharatGPT: A Suite of Generative AI Tech for India' in collaboration with BharatGen, IIT Kanpur. I have developed various algorithms based on statistical analysis for **Subgraph Querying** in large complex graphs. Some of the many areas that can benefit from the querying of subgraph structures are information extraction, recommendation systems, disease diagnostics, fraud detection. During my post-doc, I worked on **HyGraphs**, where as a team we explored development of a novel data model combining **property graphs** and **time series data models**, with an aim to provide a single platform for seamless, intuitive and efficient processing of large scale datasets which are dynamic in nature.

TECHNICAL SKILLS

Languages : C, C++, Python, Java, R, JavaScript, PHP, Shell scripting
Databases : MySQL, Neo4j
Libraries : PyTorch, Tensorflow, Scikit-learn, Numpy, Pandas
Tools : git, LaTeX, Docker, Weka

PHD THESIS

Subgraph Matching and Mining in Large Graphs

Supervisor: Arnab Bhattacharya

- Developed algorithms for Approximate Subgraph Matching in both deterministic and probabilistic graphs.
- Proposed a Graph Neural Network model for robust node embeddings with positional information.

EXPERIENCES

Project Executive Officer

BharatGen, IIT Kanpur, India

Sep 2025 - Present

- Project: 'BharatGPT: A Suite of Generative AI Tech for India'.
- Funding Agency: Department of Science and Technology (DST), Government of India

Post-doctoral Researcher

LIRIS Research Lab, CNRS, Université Claude Bernard Lyon 1, France

May 2024 - Apr 2025

- Project: 'HyGraph: Querying and Analytics on Hybrid Graphs'.
- Funding Agency: French Agence Nationale de la Recherche (ANR) and the Deutsche Forschungsgemeinschaft (DFG; German Research Foundation)

Senior Student Research Associate

IIT Kanpur, India

Aug 2022 - Dec 2023

- Project: 'Short and Long-term Fog Prediction for North Indian Cities'.
- Funding Agency: Scheme for Transformational and Advanced Research in Science (STAR), Ministry of Education, Government of India.

Summer Intern

Flipkart Internet Private Limited

Jun 2015 - Jul 2015

- Worked in a project to prepare a model for identification of 'fake' listings.

PUBLICATIONS

- "A Case Study of Cross-Lingual Zero-Shot Generalization for Classical Languages in LLMs", V S D S Mahesh Akavarapu, Hrishikesh Terdalkar, Prमित Bhattacharyya, **Shubhangi Agarwal**, Vishakha Deulgaonkar, Chaitali Dangarikar, Pralay Manna, and Arnab Bhattacharya, 63rd Annual Meeting of the Association for Computational Linguistics (ACL), 2025, Austria.
- "Towards Hybrid Graphs: Unifying Property Graphs and Time Series", Mouna Ammar, Christopher Rost, Riccardo Tommasini, **Shubhangi Agarwal**, Angela Bonifati, Petra Selmer, Evgeny Kharlamov and Erhard Rahm, 28th International Conference on Extending Database Technology (EDBT), 2025, Spain.

- “Transforming Time Series into Graphs and Back with HyGraph” , Mouna Ammar, **Shubhangi Agarwal**, Angela Bonifati and Erhard Rahm, Proceedings of the Workshops of the EDBT/ICDT Joint Conference, 2025, Spain.
- “VeNoM: Approximate Subgraph Matching with Enhanced Neighbourhood Structural Information” , **Shubhangi Agarwal**, Sourav Dutta and Arnab Bhattacharya, 7th Joint International Conference on Data Science and Management of Data (CODS-COMAD), 2024, India.
- “VerSaChI: Finding Statistically Significant Subgraph Matches using Chebyshev’s Inequality” , **Shubhangi Agarwal**, Sourav Dutta and Arnab Bhattacharya, Proceedings of the International Conference on Information and Knowledge Management (CIKM), 2021, pages 2812-2816, Australia.
- “GraphReach: Position-Aware Graph Neural Network using Reachability Estimations” , Sunil Nishad, **Shubhangi Agarwal**, Arnab Bhattacharya and Sayan Ranu, Proceedings of the International Joint Conference on Artificial Intelligence (IJCAI), 2021, pages 1527-1533, Canada.
- “ChiSeL: Graph Similarity Search using Chi-Squared Statistics in Large Probabilistic Graphs” , **Shubhangi Agarwal**, Sourav Dutta and Arnab Bhattacharya, Proceedings of the International Conference on Very Large Data Bases (VLDB), 2020, pages 1654-1668, Japan.

EDUCATION

Ph.D. in Computer Science and Engineering CGPA: 8.25
Indian Institute of Technology Kanpur, Uttar Pradesh, India 2014 - 2023

Bachelor of Technology Computer Engineering CGPA: 8.71
Sardar Vallabhbhai National Institute of Technology, Surat, Gujarat, India 2010 - 2014

TEACHING

Instructor Sep 2024 – Dec 2024
Université Claude Bernard Lyon 1, France

- Program: M2 - Data and Intelligence for Smart Systems (DISS)
- Course: Data Processing and Analytics (Jointly with others)

Teaching Assistant Aug 2014 - Apr 2021
IIT Kanpur, India

- Graded and evaluated projects for various courses of Computer Science.

Senior Tutor Aug 2017 - Apr 2018, Aug 2019 - Apr 2020
IIT Kanpur, India

- Led teams of strength ~60; Assisted in backend management, paper-setting and grading.

Teaching Assistant for MOOC July 2017 - Sep 2017
NPTEL - Remote

- Crafted objective questions and resolved student queries on Fundamentals of Database Systems.

ADMINISTRATIVE EXPERIENCES

Program Committee Member 2025 - current

- GRADES-NDA Workshop - SIGMOD/ PODS (2025), ICWE (2025), TGD Workshop - EDBT/ ICDT (2025).

External Reviewer 2020 - current

- WSDM (2024), CIKM (2021, 2022), DASFAA (2022), CoDS-COMAD (2020, 2021), KDD (2021).